

Lecture Topic	Slides Filename	Open Question
Intro AI-ML, prob	Lecture 2 - Feb24-27 2025 -IntroAI-IntroML-IntroProb.pdf	List and briefly describe the three main ML paradigms
		Describe the ML paradigm of supervised learning and give an example of supervised learning algorithm presented during the course
		Describe the difference between supervised and unsupervised ML algorithms
		Describe the ML paradigm of unsupervised learning and give an example of unsupervised learning algorithm presented during the course
		Describe what a classification problem is, and make a simple binary classification example
		Describe the Bayes' Rule, its elements, and why is it useful. Name a classification algorithm whose formulation exploits the Bayes' Rule.
		Illustrate the Naïve Bayes Classifier and its main assumption.

		Explain the Bias/Variance Decomposition Theorem seen during the course. No proof required.
Labs overview (data preprocessing, pipeline, cross validation...)	Lecture - Feb27 - How to build a pipeline.pdf	Describe the key stages involved in a typical deep learning project workflow, from initial problem understanding to having a usable model. For each stage, briefly explain its main purpose.
		Before diving into data preprocessing for a deep learning project, explain three critical characteristics or potential issues related to your dataset that you should investigate.
		Explain why raw data, particularly images, generally requires preprocessing before being used to train a deep learning model. Describe at least two distinct common preprocessing steps and one data augmentation technique, outlining the specific purpose or benefit of each.
		Explain the rationale behind splitting a dataset into training, validation, and test sets in a machine learning project. Describe the distinct role and purpose of each of these sets.
		Describe the iterative process of training a deep learning model. Explain what fundamentally occurs during this training phase. What is the ultimate aim of training?
		What are hyperparameters in the context of designing and training deep learning models? Provide and discuss three distinct examples of common hyperparameters.

		The process of developing a successful deep learning model is often described as highly iterative, involving continuous "improving through experimentation." Explain what this "iterative nature" means in the practical context of a deep learning project.
		Explain why it would be highly problematic to directly use the test set for hyperparameter tuning. How can this issue be avoided when building a ML pipeline?
Learning Theory + Unsupervised Learning	Lecture 3 - Mar3 2025 - LearningTheory+UnsupervisedLearning.pdf	What is a loss function and why is its choice crucial when tackling a Machine Learning problem?
		Explain the concepts of True Risk and Empirical Risk. What is the goal of a learning algorithm in relation to them?
		Describe the core idea behind Probably Approximately Correct (PAC) learning. What do the "Probably" and "Approximately Correct" aspects refer to, and how do the parameters epsilon (ϵ) and delta (δ) relate to these concepts?
		Describe the Vapnik-Chervonenkis (VC) dimension of a hypothesis class. Explain the concept of "shattering" a set of points and how it is used to determine the VC dimension.
		Describe the phenomena of overfitting and underfitting. What does it mean for a model to overfit the training data, and how does this typically affect its performance on previously unseen examples?

		Explain the K-fold Cross Validation procedure for model selection. For which data regime (i.e., training set size) can it be particularly useful?
		Illustrate the key principles and step-by-step functioning of the Linear Regression algorithm.
		Illustrate the key principles and step-by-step functioning of the k-Means clustering algorithm.
		Illustrate the key principles and step-by-step functioning of the Gaussian Mixture Model.
		Illustrate the key principles and step-by-step functioning of the Expectation-Maximization (EM) algorithm.
		Illustrate the key principles and step-by-step functioning of the k-Nearest Neighbors classification algorithm.
		Describe the main advantages and disadvantages of the k-Nearest Neighbors (k-NN) algorithm. Explain why it is considered a non-parametric method and discuss its sensitivity to the choice of 'k' and feature scaling.

k-NN, Perceptron, Kernels, SVM, MLP	Lectures 4 + 5 - Mar6 2025 KNN-Perceptron-kernels-MLP-SVM.pdf	Explain the Perceptron learning algorithm for binary classification. Describe its main components and the process of weights updating.
		What is the "Kernel Trick" in machine learning? Explain its purpose, particularly how it allows linear algorithms (like the Perceptron or SVM) to learn non-linear decision boundaries.
		Describe the basic architecture of a Multi-Layer Perceptron (MLP). Explain the role of hidden layers and non-linear activation functions in enabling MLPs to model complex relationships, overcoming the limitations of the single-layer Perceptron.
		Explain the core idea behind the Support Vector Machine (SVM) for linearly separable data (Hard Margin SVM). Describe what the "margin" is and why SVM aims to maximize it.
		Explain why the distance measure choice is crucial for the kNN algorithm. Make at least 3 examples of metrics.
		What is an Artificial Neural Network?
		Describe the Gradient Descent algorithm. How is it usually employed in Machine Learning?

Neural Networks, Backpropagation & Optimization	lecture-10-03-25-neural networks- optimization.pptx.pdf	What is the difference between the Gradient Descent and the Stochastic Gradient Descent algorithms?
		Describe the Stochastic Gradient Descent algorithm
		Describe the backpropagation algorithm
		Describe the concepts of batches, epochs, and learning rate within the context of optimization algorithms for ML
		Explain and describe the concept of Momentum in the context of optimization methods for learning models.
		Describe the ADAM algorithm
		Illustrate the key principles and step-by-step functioning of the Logistic Regression algorithm for binary classification.

Logistic Regression & Calibration	Lecture 13mar2025 - regression-grad-descent+calibration.pdf	Some classification methods also output a score that can be interpreted as a probability of predicted class membership. Define the property of a classifier being 'well-calibrated' with respect to its probabilistic outputs. Explain why this is an important property in certain domains and name a simple application example.
		A bank wants to predict whether a loan applicant is likely to default ($Y=1$) or not default ($Y=0$) on their loan. They have historical data including features like applicant income, credit score, loan amount, and employment duration, along with the actual outcome (defaulted or not). Explain which method between Logistic Regression or Linear Regression would be most suitable for this task and why.
		Explain the main problems or limitations of directly using standard Linear Regression for this binary classification task. How does Logistic Regression address these limitations, particularly concerning the range and interpretation of its output?
		Illustrate the Expected Calibration Error metric
		Name and briefly describe one post-training classifier calibration method seen in class
		What is a convolutional neural network? Describe the overall architecture and its main components, with a particular focus on convolutional layers
		What is a convolutional neural network? Describe the overall architecture and its main components, with a particular focus on pooling layers

Introduction to Deep Learning (DL - part 1B)	lecture-17-03-25-IntroDeepLearning-CNN.pptx.pdf lecture - 20-03-25-IntroDLContd.pptx.pdf	What is a convolutional neural network? Describe the overall architecture and its main components, with a particular focus on activation functions.
		What is a convolutional neural network? Describe the overall architecture and its main components, describe the training procedures and strategies with a particular focus on data preprocessing and weight initialization
		What is a convolutional neural network? Describe the overall architecture and its main components, describe the training procedures and strategies with a particular focus on batch normalization
		What is a convolutional neural network? Describe the overall architecture and its main components, describe the training procedures and strategies with a particular focus on transfer learning
		What is a convolutional neural network? Describe the overall architecture and its main components, describe the training procedures and strategies with a particular focus on dropout regularization
		Describe the AlexNet architecture
		Describe the VGG network architecture

Deep Learning Architectures: Alexnet, VGG, ResNet, RNNs	lecture - 24-03-25-DL-Architectures-Visualizations.pdf	Describe the Residual Network (ResNet) architecture
		Describe the Recurrent Neural Network (RNN) architecture
		Considering the AlexNet architecture, explain how the choice of the ReLU activation function and the use of dropout, combined with data augmentation, allowed to successfully train such a deep model on the large-scale ImageNet dataset. Which problems do these components address, and why were they particularly important for overcoming training difficulties?
		The VGG network demonstrated that significantly increasing network depth could lead to better performance on complex tasks like ImageNet classification. A key aspect of VGG was its highly uniform architecture, primarily relying on repeating stacks of small 3x3 convolutional filters and 2x2 pooling layers. What did the success of VGG suggest about the relationship between network depth and the ability to learn powerful visual representations?
		GoogLeNet (with its Inception modules) aimed to create deeper networks while being more computationally efficient. Explain the core idea behind the Inception module. Discuss the computational challenges this approach initially faced and how the use of 1x1 convolutions ("bottlenecks") helped manage this complexity and reduce the number of parameters.
		As networks got deeper post-VGG, researchers observed the "degradation" problem where simply stacking more layers hurt performance, even on the training set, indicating an optimization challenge rather than just overfitting. Explain this phenomenon. Describe the fundamental innovation introduced by ResNet and intuitively explain how this mechanism helps overcome the optimization difficulty and enables the training of much deeper networks more effectively.
		What is an LSTM?

From RNNs to Transformers	lecture - 27-03-25-RNN-Transformers.pdf	What are the main differences between an RNN and a Transformer?
		Discuss the concept of attention and its implementation (general attention layer) in the context of RNNs and Transformers
		Describe attention layers and self attention layers, discussing similarities and differences
		Describe self attention layers and their masked and multi-head versions
		Describe Transformers and their main components, with special focus on the encoder block
		Describe Transformers and their main components, with special focus on the decoder block